What Predicts Where Maine Builds?

Municipal characteristics and the geography of housing permits

CS 570 – Project 1

1 Introduction

Maine has a housing shortage, and the state has only recently started requiring towns to report their building permits every year. That makes permits the earliest hard number we have on where new housing is actually being built. Our question is a narrow one: which municipal characteristics go with a higher rate of new construction?

We did not start there. The first target we tried was the total number of permits a town issues, and it turned out to be a disappointing one. Permit counts track population almost mechanically, so a model of counts mostly recovers how big a town is, which we already knew from the population column. We switched to the permit rate, permits issued per 1,000 residents, and kept the same candidate predictors: median household income, median home value, population density, the owner-occupied share, the multifamily share of the housing stock, and the median year the stock was built.

The short version of what we found is that home value matters a great deal, income does not, and density runs the opposite way from what we expected. On top of population, the characteristics roughly double the share of variance explained, reaching an R-squared of about 0.25 out of fold, and none of the flexible models we tried buys anything over a plain linear model. A smaller second task puts a different question to the same towns: which of them permitted any multifamily housing at all?

2 Dataset

Permit counts come from the 2025 Statewide Housing Data workbook, which combines a municipal survey, records from the Land Use Planning Commission, and Census Building Permits Survey estimates for the 220 municipalities that did not respond to the survey. The characteristics come from the 2020-2024 American Community Survey five-year estimates at the county subdivision level, joined to the Census Gazetteer for land area. We matched 528 of the 529 workbook municipalities to an ACS subdivision; the exception is the aggregate UNORGANIZED TERRITORIES row, which is not a single place.

Table 1 lists the variables we use.

Table 1: Variables used in the analysis.

Variable	Source	Definition
Permit rate	workbook	Permits issued in 2024 per 1,000 residents
log population	ACS B01003	Total population (log)
log income	ACS B19013	Median household income, dollars (log)
log home value	ACS B25077	Median value of owner-occupied units, dollars (log)
log density	ACS + Gazetteer	Residents per square mile of land (log)
Owner share	ACS B25003	Share of occupied units that are owner-occupied
Multifamily share	ACS B25024	Share of housing units in buildings with 2+ units
Median year built	ACS B25035	Median year the housing stock was built

Two features of the permits are worth spelling out, because they shape everything after this. First, a town

reports through one of two channels, a local survey or Census estimates, and the two do not agree exactly. Across the 169 towns that appear in both, the local totals run about 8% above the Census estimate, so we keep the reporting channel as a predictor rather than pretending the two series are interchangeable. Second, the missingness is structural rather than random. Accessory dwelling units and affordable-unit counts exist only for the 182 towns that filled in the survey, the Census columns exist wherever the Census publishes them, and 90 municipalities report nothing at all. Dropping the non-reporting towns and one unorganized-territory row with no population leaves 438 towns with a permit count, and 419 of those also have the ACS characteristics. That 419 is the sample used below.



Figure 1: Permit rates across the 419 towns. (a) Permits per 1,000 residents, x-axis truncated at 40 (15% of towns are at zero). (b) Median rate by home-value quintile. (c) Median rate by density quintile.

Permit rates vary a lot from town to town. The median town issued 4.8 permits per 1,000 residents, but the upper quartile is 9.2, the maximum is 333, and 15% of towns issued none at all. Panel (b) shows the first pattern we looked at: the median rate climbs with housing wealth, from 1.9 permits per 1,000 in the cheapest quintile to 6.7 in the fourth, then drops back a little. Panel (c) goes the other way, which surprised us. The sparsest quintile has the highest rate (9.2), and the other four are close to five.

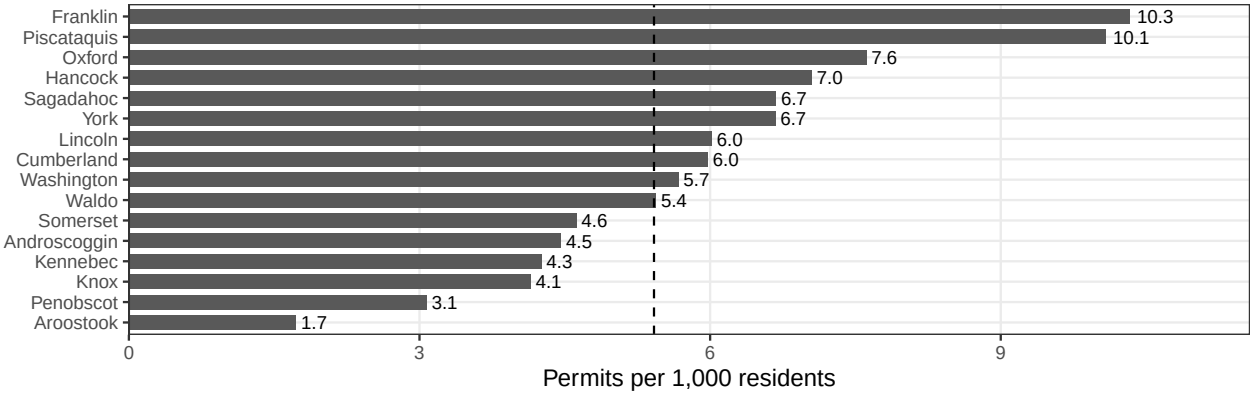


Figure 2: Permit intensity by county. Bars are aggregate permits per 1,000 residents; the dashed line is the statewide rate. Counties are ordered by intensity.

Aggregate rates differ by a factor of six between counties, from 1.7 permits per 1,000 residents in Aroostook to 10.3 in Franklin. That ranking is not the one we would have guessed. The top of the list is rural and inland, and the two most populous counties sit in the middle, which is part of why county stays in the model as a control rather than as a stand-in for size.

3 Methods

Work was done in R with tidymodels; the preprocessing is in `R/data_prep.R` and the models in `R/models.R`. The response is log(1 + permits per 1,000 residents). Rates cannot be negative, they are skewed hard to the right, and 64 towns issued nothing at all, so we add one before taking logs. On that scale an RMSE of 0.8 is roughly a factor of $e^{0.8}$ in the raw rate, which is the right way to read the tables below.

We fit each task as a ladder of models, holding the recipe and the resampling fixed so that the comparison is fair:

- Size only: OLS of the rate on log population.
- Size plus county: the same model with 15 county dummies.
- Characteristics: the seven predictors, no county.
- Characteristics plus county: the full specification.
- Ridge and lasso: the same design, with the penalty tuned over 20 values.
- Random forest (`ranger`), tuning `mtry` and `min_n`.
- XGBoost, tuning depth, learning rate and `min_n`.

The second task repeats the ladder with a binary outcome, whether the town permitted any multifamily unit at all, and scores models by AUC instead of RMSE.

We expected collinearity to be a problem and it was: population and density correlate at 0.91, income and home value at 0.65, and variance inflation factors reach 8.2 and 7.4. Ridge and lasso are in the ladder as a check on how much of the county effect is genuine rather than an artifact of that overlap.

Every model is scored by repeated 10-fold cross-validation with three repetitions, with folds stratified on the outcome for the multifamily task. Tuning happens inside each resample, so the numbers reported are out of fold and no model has seen its own test rows. Fits are cached in `results/*.rds`, so the report can be rebuilt without refitting.

The tuning settled on small models: a ridge penalty of 0.0886, a lasso penalty of 0.0264, a forest with `mtry` 2 and `min_n` 2, and boosting at depth 2 with a learning rate of 0.005.

4 Results

4.1 Permit rate

Table 2: Cross-validated performance for the permit-rate regression. The response is log(1 + permits per 1,000); standard errors are over the 30 resamples; the last column compares RMSE with the full OLS model.

Model	RMSE (SE)	MAE (SE)	R2 (SE)	vs OLS
Size only (OLS)	1.073 (0.022)	0.824 (0.018)	0.069 (0.011)	+12.1%
Size + county (OLS)	0.981 (0.021)	0.755 (0.015)	0.212 (0.026)	+2.5%
Characteristics (OLS)	0.983 (0.018)	0.749 (0.014)	0.201 (0.023)	+2.7%
Characteristics + county (OLS)	0.957 (0.019)	0.734 (0.012)	0.249 (0.026)	–
Ridge	0.951 (0.018)	0.729 (0.013)	0.251 (0.025)	-0.7%
Lasso	0.954 (0.018)	0.725 (0.014)	0.244 (0.024)	-0.3%
Random forest	0.959 (0.019)	0.718 (0.015)	0.271 (0.030)	+0.2%
XGBoost	0.983 (0.026)	0.722 (0.015)	0.236 (0.028)	+2.7%

Population on its own explains 7% of the out-of-fold variance. Adding the seven characteristics takes that to 20%, and adding the county dummies as well reaches 25%, so the characteristics roughly double what population buys and county is worth another few points on top.

The flexible models do not improve on this. The random forest has the best R^2 (0.271), but its RMSE sits inside one standard error of the OLS model, and XGBoost comes out slightly worse than OLS. Ridge collapses back to OLS (the selected penalty is essentially zero) and lasso keeps most of the county dummies at a small penalty, which says the county effects are not an artifact of the dummy expansion. Our reading is that on the log scale the relationship is close to additive, and a quarter of the variance is roughly the ceiling for this feature set.

Table 3: Standardised OLS coefficients for the permit rate, per one standard deviation change in each predictor, with 95% intervals. County effects are omitted here but included in the fitted model.

Variable	Std. coef.	95% CI	p value
log population	-0.289	-0.54 to -0.04	0.025
log median household income	-0.085	-0.22 to +0.05	0.230
log median home value	+0.450	+0.29 to +0.61	< 0.001
log population density	-0.284	-0.56 to -0.01	0.041
owner-occupied share	+0.052	-0.10 to +0.20	0.492
multifamily share of stock	+0.187	+0.03 to +0.34	0.017
median year built	+0.022	-0.09 to +0.13	0.697

Home value is the main story. One standard deviation of median home value goes with a 0.45 standard deviation increase in the log permit rate, and in the unstandardised model a doubling of home value corresponds to roughly a 1.98-fold higher rate. Income takes the wrong sign (-0.09) and cannot be separated from zero once home value is in the model, so we do not read anything into it. Population and density are both negative and are hard to pull apart; the plain way to say it is that, conditional on county and on each other, larger and denser towns permit less per resident. The multifamily share of the existing stock is positive, which fits the obvious explanation that towns which already have apartments keep building them.

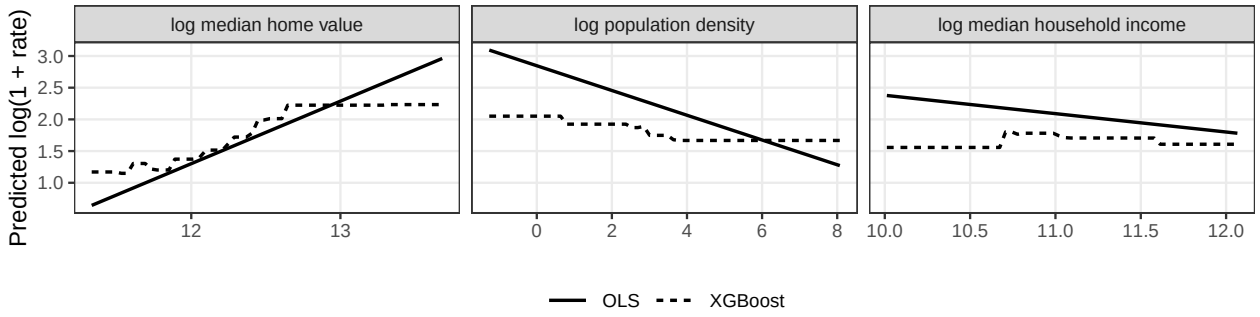


Figure 3: Partial dependence of the permit rate on three predictors, from the OLS model (solid) and XGBoost (dashed). Each panel varies one predictor and holds the others at their observed values.

Figure 3 is the same result from the other side. The XGBoost curves follow the directions of the OLS slopes, but they flatten in the middle of the range and bend at the ends, where there are fewer towns to learn from. Home value is the exception and stays monotone over almost its whole range. Density is flat above about 100 residents per square mile and rises only among the emptiest towns.

We checked the fit two ways. The zeros were the first worry: refitting on the 355 towns with a strictly positive rate, using a plain log(rate) response, leaves the coefficients in the same direction and about the same size (home value 0.72 against 0.99, population -0.39, density -0.24, multifamily share 2.00), so the zero towns are not driving the result. The second is the reporting channel, discussed above; keeping it in the model means the coefficients are conditional on who counted the permits.

4.2 Multifamily housing

The outcome here is binary, so the same predictors and the same resampling are scored by AUC rather than RMSE.

Table 4: Cross-validated performance for the multifamily classification (mf_any). Sensitivity and specificity are at the 0.5 threshold.

Model	AUC (SE)	Accuracy (SE)	Sensitivity	Specificity
Size only (logistic)	0.881 (0.013)	0.869 (0.008)	0.96	0.50
Logistic regression	0.874 (0.017)	0.869 (0.009)	0.95	0.54
Penalised logistic	0.887 (0.013)	0.869 (0.009)	0.96	0.48
Random forest	0.868 (0.013)	0.886 (0.007)	0.96	0.57
XGBoost	0.870 (0.013)	0.877 (0.007)	0.94	0.62

Telling apart towns that permitted multifamily housing is much easier than explaining the rate; every model lands around AUC 0.88. What interested us is that the characteristics add nothing. Logistic regression on log population alone gets 0.881, and the best of the four full models, Penalised logistic, gets 0.887, a difference well inside the noise. Whether a town permits any apartments at all is mostly a question of how large it is.

5 Discussion

Housing wealth, not household income, is what tracks where Maine builds. That is the finding we would lead with, and it is not what we expected going in; income is the variable everyone reaches for first. The two correlate at 0.65, so the pair is best read together: what looks like a market signal is the value of what is already there, not the earning power of the people who live there.

Density running negative is the second finding, and the one we trust least, because population and density are nearly the same variable in a state with this settlement pattern. We would not want to claim that denser towns build less; the safer claim is that the emptiest fifth of towns builds at roughly double the median rate, which is consistent with construction on cheap land at the edge of service centers.

The third finding is the ceiling. Even the forest, with every characteristic and 15 county dummies, explains about a quarter of the out-of-fold variance in the log rate, and the flexible models do not beat the additive one. That tells us the missing variables are not model capacity; they are zoning, sewer and water capacity, buildable land, local politics and the age of the population, none of which the ACS publishes at this geography. A quarter of the variance is a real result about how much of local permitting is municipal socio-economics rather than a modeling failure.

Two limitations we should flag. The sample drops the 90 towns that do not report, and those are the smallest places, so what we describe is the permitting behavior of towns that build and report rather than of the state as a whole; the 20-odd towns with suppressed income or home value sit in the same corner of the sample. And this is a single cross-section, so home value is as much an outcome of scarcity as a cause of construction, and the coefficients are descriptive contrasts rather than effects.

If we did this again we would start from the rate rather than the count, since we spent a while modeling a target that turned out to be mostly population. The obvious next step is a panel: the state has only a few years of annual reporting so far, but once there are more, town fixed effects would absorb the local politics that currently sit in the error term.